

(f) Scatter diagrams and correlation

Students should be taught to:

plot data as points on a scatter diagram;

recognise positive, negative and zero correlation by eye;

understand the distinction between correlation, causality and a non-linear relationship;

fit a line of best fit passing through $(\bar{x}, \bar{y})$ to the points on a scatter diagram, by eye may be required;

understand the pitfalls of interpolation and extrapolation;

interpret data presented in the form of a scatter diagram.

The labelling and scaling of axes may be required.

Terms such as strong or weak will be expected.

The points lying on the circumference of a circle are related but show zero correlation.

Questions will state when $(\bar{x}, \bar{y})$ is required.

Particularly the problem of extrapolating beyond the range.